

CAROL JACOB

Karunya Institute of Technology and Sciences, Coimbatore, Tamil Nadu, India

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PROFESSIONAL SUMMARY

Dynamic individual with a strong foundation in biosciences and technology, complemented by hands-on laboratory experience. Detail-oriented and innovative, with proven leadership skills and a commitment to impactful research. Eager to contribute to biotechnology projects that benefit society.

EDUCATION

GRADUATION-B.TECH. BIOTECHNOLOGY

KARUNYA INSTITUTE OF TECHNOLOGY AND SCIENCES	GPA : 8.2	2022-2026
CLASS 12	SCORED : 93.6%	2021
KENDRIYA VIDYALAYA NO.2 AIR FORCE STATION JODHPUR		
CLASS 10	SCORED : 95.4%	2019
KENDRIYA VIDYALAYA NO.1 AIR FORCE STATION GORAKHPUR		

SKILLS

- Phase Contrast Microscopy - Aseptic Techniques - Stem Cell Culture - Microbiology Methods & Antibiotic Testing Cell - Cell Culture Maintenance	- Leadership & Teamwork Event - Hosting & Science Outreach - Biomaterials & Biocompatibility - Scientific Writing & Communication - Basic Bioinformatics Concepts
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CERTIFICATION COURSES

- AI in healthcare - Introduction to Medical Biochemistry - Biomaterials Biocompatibility - Molecular Biology and Biotechnology Principles - Biomaterials for bioprinting - Preclinical Basics: Toxicity - Bioimaging & Literature - Cell Signalling: Communication at the Molecular Level	- Pharmaceutical Project Management - Structural Biology - Diploma in Drug Development - Cardiovascular Pharmacology - Vaccines and the Immune System - Understanding Molecular Biology - Basics of Molecular Genetics - The Fundamentals of Chromatography
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SCHOLARSHIPS

- Institutional Merit Scholarship 2022
- Institutional Merit Scholarship 2024

EXPERIENCE

Student Intern - Medipulse Hospital, Jodhpur, Rajasthan	2023
Gained insights into diagnostic tests and laboratory protocols by observing key processes in the pathology lab, molecular lab, and blood bank.	
Medical Microbiology Intern - Dr. B.Lal Institute of Biotechnology, Jaipur, Rajasthan	2024
Participated in microbiological techniques such as microbial isolation and antibiotic sensitivity testing, enhancing skills in diagnostic methodologies.	
Molecular Biology - Xcellogen Biotech, Trivandrum, Kerala	2024
Observed and practiced techniques like DNA extraction and PCR, improving skills in molecular diagnostics and experimental procedures relevant to plant biology.	
Molecular Genetics Intern – Cochin University of Science and Technology, Kochi, Kerala	2025
Trained in phage titration, zebrafish gut microbiota isolation, genome annotation and observed FACS and RT-PCR.	
Research Trainee – Stem Cell Research Laboratory, Karunya Institute of Technology and Sciences	2025
Learned subculturing, cryopreservation, and mesenchymal stem cell isolation; strengthened aseptic and handling skills.	
Currently working on HeLa cell line, L929 Mouse Fibroblasts and A549 Lung Cancer cell lines to develop an anti-cancer property rich hydrogel from grape phytochemicals.	

PROJECTS

- Health and Wellness Tracking Application using C Programming
- Microbial Footprints - Investigating Microorganisms on Animal Feet
- Nanoparticle Solutions for Energy Efficient Saltwater to Freshwater Transformation
- Enzyme Kinetics Analysis For Enhanced Lactic Acid Production Via Solid-State Fermentation
- In Silico Analysis, Extraction And Characterization Of Polyphenols From *Phyllanthus emblica* Bark
- Smart Escape Routes: Utilizing IoT And AI Networks For Toxic Gas Mitigation In Mining Industry
- Development of Biopaper and Bioplastic Products using Natural Polymer Sources
- In Silico Evaluation of Grape Polyphenols as Multi-Target Activators using Bioinformatics

CONFERENCES AND WORKSHOPS

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| • EVOGEN: National Level Technical Symposium | 2023 |
| • INSPIRE: A Premier Industry Academia Student Conference | 2024 |
| • NEUROSCAPE: International Biotech Conference on Insights and Innovations Navigating the Neuroscape | 2024 |
| • MINDKRAFT: Annual Techno-Management Fest | 2024 |
| • WOERCST: Workshop on Opportunities in Educational & Research Collaborations in Taiwan | 2024 |
| • Workshop on Transformative Healthcare Insights through Artificial Intelligence & Machine Learning | 2024 |
| • Workshop on Chicken IgY Antibody and Advanced Microscopic Techniques | 2024 |
| • RT-PCR Workshop on Gene Expression Profiling | 2025 |
| • Hands-on training in Animal Cell Culture Techniques, Stem Cell Research Laboratory, Karunya | 2025 |
| • Workshop on Pedigree Protein Pack | 2025 |
| • ICFAST: International Conference on the Frontier Areas of Science and Technology, NIT Calicut | 2025 |
| • RBAT-X-2026: International Seminar on Recent Biochemical Approaches in Therapeutics, University of Kerala | 2026 |

POSTER PRESENTATIONS

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| • Teratogenicity and Pregnancy: Understanding the Impact of Adverse Drug Reactions | 2024 |
| • Smart Escape Route System: Utilizing IoT and AI Networks for Toxic Gas Mitigation in the Mining Industry | 2024 |
| • DNA Repair Mechanisms: Exploring the Potential of Nutraceuticals | 2024 |
| • Immunofluorescent Biomarker Imaging | 2024 |
| • Analysis of Inheritance Patterns in <i>Pisum sativum</i> through Dihybrid Cross | 2024 |
| • Chimerism in Hematopoietic Stem Cell Therapeutics | 2024 |
| • Phytochemical Profiling and Functional Evaluation of Grape Leaf Extracts in Comparison with Resveratrol: Antioxidant, Antibacterial, and Cytotoxic Activities in L929 Fibroblast Cells | 2025 |
| • Phytochemical Profiling and In-Vitro Anticancer Evaluation of Red Globe (<i>Vitis vinifera</i>) Grape Leaf Extracts in A549 Lung Cancer Cells | 2026 |

AWARDS AND RECOGNITIONS

She Icon Award – Women’s Excellence Awards 2026

Global Women Leadership Summit, Karunya Deemed University, January 2026

Recognized for exemplary character, leadership potential, ethical conduct, and resilience.

EXTRA CURRICULARS AND OUTREACH

President, Student Association, Division of Biotechnology

Appointed as the President of the Student Association, Division of Biotechnology effective from August 2025

National Cadet Corps, 5 (Tamil Nadu) Girls Battalion

Enrolled as a cadet; appointed **Company Quarter Master Sergeant** in August 2023 and promoted to the highest rank titled **Senior Under Officer** in August 2024, leading more than 100 cadets during camps and parades to date.

Participated in training camps at Bannari Amman Institute of Technology (2023) and KPR Institute of Engineering and Technology (2024), enhancing military skills, and completed an Army Attachment Camp at the Madras Regimental Centre, Wellington. Achieved ‘**A**’ **grade** in NCC- A and B Certificate Examination 2024

Competitions and Recognitions

Achieved recognition in Best Cadet Competitions, secured First Position in Glider Making during Mindkraft 2023, and selected for an Internal Panel Presentation through MSME, a GOI initiative for a project on IoT and AI networks.

Nominated as Gender Champion for the Academic Term 2023-24 at the core committee of the university.

Overall third rank with a CGPA of 9.54 in the first semester of BTech Biotechnology.

Event Hosting and Coordination

Compered key events like the Republic Day Parade, Civil Services Conclave, and Department Investiture Ceremony. Conducted social service events like old age home visits, tree plantation drives, blood donation drives, and women empowerment programs, school biotechnology exposure programs and participated in various government initiatives conducted in the university.